CHEN Chaofeng (陈超锋)

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RESEARCH INTERESTS

Computer vision and image processing, with a focus on image/video quality assessment, image restoration and enhancement, and multi-modality generative models.

RESEARCH EXPERIENCE

Tenure Track Assistant Professor Present
School of Artificial Intelligence, Wuhan University

Postdoctoral Research Fellow Sep. 2021 - Jan. 2025
S-Lab, Nanyang Technological University; worked with Prof. Weisi Lin and Prof. Kelly

EDUCATION

The University of Hong Kong (HKU) - Ph.D. in Computer Science Sep. 2015 - Jan. 2021

Huazhong University of Science and Technology (HUST) - B.Eng. in Computer Science Sep. 2011 - Jun. 2015

SELECTED PUBLICATIONS

Full list: Google Scholar. * Equal contribution; # Corresponding author.

Recent Publications (2025-2026)

[1] **[MM 2026]** Lu Gan, Hanyu Yan, **Chaofeng Chen**, Junqi Hu, Dan Zeng. GeoMAR: Unleashing Geometrically Aligned Features for Masked Autoregressive Blind Face Restoration.

[2] **[MM 2026]** Kexin Ma, Jing Xiao, **Chaofeng Chen**, Geyong Min, Guibo Zhu, Jinqiao Wang, Liang Liao. Decoupled Similarity for Task-Aware Token Pruning in Large Vision-Language Models.

[3] **[MM 2026; Oral]** Mengtian Li, Xinru Guo, Xiaoru Lin, Xiao Rong, Zhifeng Xie, **Chaofeng Chen**[#]. ETPDesigner: Multi-Agent Orchestration for Interactive Multimodal Electronic Theater Program.

[4] **[TPAMI 2026]** Mengcheng Lan, **Chaofeng Chen**, Jiaxing Xu, Zongrui Li, Yiping Ke, Xudong Jiang, Yingchen Yu, Yunqing Zhao, Song Bai. Text4Seg++: Advancing Image Segmentation via Generative Language Modeling.

[5] **[TMM 2026]** Wen Sun, **Chaofeng Chen**, Liang Liao, Weisi Lin. Prompt-Image-Caption Consistency for AI-Generated Image Quality Assessment.

[6] **[TMM 2025]** Hui Li, Liang Liao, **Chaofeng Chen**, Xiaopeng Fan, Wangmeng Zuo, Weisi Lin. Denoised Semantic Features for Local Consistent No-Reference Image Quality Assessment.

[7] **[ICCV 2025; Highlight]** Yachun Mi, Yu Li, Weicheng Meng, **Chaofeng Chen**, Chen Hui, Shaohui Liu. MVQA: Mamba with Unified Sampling for Efficient Video Quality Assessment.

[8] **[ICLR 2025]** Mengcheng Lan, **Chaofeng Chen**, Yue Zhou, Jiaxing Xu, Yiping Ke, Xinjiang Wang, Litong Feng, Wayne Zhang. Text4Seg: Reimagining Image Segmentation as Text Generation.

Selected Conference Papers

[9] **[MM 2024; Oral] Chaofeng Chen**, Yang Sensen, Haoning Wu, Liang Liao, Zicheng Zhang, Annan Wang, Wenxiu Sun, Qiong Yan, Weisi Lin. Q-Ground: Image Quality Grounding with Large Multi-modality Models.

[10] **[ECCV 2024] Chaofeng Chen**[*], Annan Wang[*], Haoning Wu, Liang Liao, Wenxiu Sun, Qiong Yan, Weisi Lin[#]. Enhancing Diffusion Models with Text-Encoder Reinforcement Learning.

[11] **[AAAI 2024] Chaofeng Chen**, Shangchen Zhou, Liang Liao, Haoning Wu, Wenxiu Sun, Qiong Yan, Weisi Lin. Iterative Token Evaluation and Refinement for Real-World Super-Resolution.

[12] **[ICML 2024]** Haoning Wu[*], Zicheng Zhang[*], Weixia Zhang, **Chaofeng Chen**, Liang Liao, Chunyi Li, Yixuan Gao, Annan Wang, Erli Zhang, Wenxiu Sun, Qiong Yan, Xiongkuo Min, Guangtao Zhai, Weisi Lin[#]. Q-Align: Teaching LMMs for Visual Scoring via Discrete Text-Defined Levels.

[13] **[ECCV 2024]** Mengcheng Lan, **Chaofeng Chen**, Yiping Ke, Xinjiang Wang, Litong Feng, Wayne Zhang. ClearCLIP: Decomposing CLIP Representations for Dense Vision-Language Inference.

[14] **[ECCV 2024]** Mengcheng Lan, **Chaofeng Chen**, Yiping Ke, Xinjiang Wang, Litong Feng, Wayne Zhang. ProxyCLIP: Proxy Attention Improves CLIP for Open-Vocabulary Segmentation.

[15] **[CVPR 2024]** Kangmin Xu, Liang Liao, Jing Xiao, **Chaofeng Chen**, Haoning Wu, Qiong Yan, Weisi Lin. Local Distortion Aware Efficient Transformer Adaptation for Image Quality Assessment.

[16] **[CVPR 2024]** Haoning Wu[*], Zicheng Zhang[*], Erli Zhang[*], **Chaofeng Chen**, Liang Liao, Annan Wang, Kaixin Xu, Chunyi Li, Jingwen Hou, Guangtao Zhai, Geng Xue, Wenxiu Sun, Qiong Yan, Weisi Lin[#]. Q-Instruct: Improving Low-level Visual Abilities for Multi-modality Foundation Models.

[17] **[ICLR 2024; Spotlight]** Haoning Wu[*], Zicheng Zhang[*], Erli Zhang[*], **Chaofeng Chen**, Liang Liao, Annan Wang, Chunyi Li, Wenxiu Sun, Qiong Yan, Guangtao Zhai, Weisi Lin[#]. Q-Bench: A Benchmark for General-Purpose Foundation Models on Low-level Vision.

[18] **[ICCV 2023]** Haoning Wu, Erli Zhang, Liang Liao, **Chaofeng Chen**, Jingwen Hou, Annan Wang, Wenxiu Sun, Qiong Yan, Weisi Lin. Exploring Video Quality Assessment on User Generated Contents from Aesthetic and Technical Perspectives.

[19] **[AAAI 2023]** Shuliang Ning, Mengcheng Lan, Yanran Li, **Chaofeng Chen**, Qian Chen, Xunlai Chen, Xiaoguang Han, Shuguang Cui. MIMO Is All You Need: A Strong Multi-In-Multi-Out Baseline for Video Prediction.

[20] **[MM 2022; Oral] Chaofeng Chen**[*], Xinyu Shi[*], Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, Shihui Guo. Real-World Blind Super-Resolution via Feature Matching with Implicit High-Resolution Priors.

[21] **[NeurIPS 2022]** Wenqi Yang, Guanying Chen, **Chaofeng Chen**, Zhenfang Chen, Kwan-Yee K. Wong. S3-NeRF: Neural Reflectance Field from Shading and Shadow under a Single Viewpoint.

[22] **[ECCV 2022]** Haoning Wu, **Chaofeng Chen**, Jingwen Hou, Liang Liao, Annan Wang, Wenxiu Sun, Qiong Yan, Weisi Lin. FAST-VQA: Efficient End-to-end Video Quality Assessment with Fragment Sampling.

[23] **[ECCV 2022]** Wenqi Yang, Guanying Chen, **Chaofeng Chen**, Zhenfang Chen, Kwan-Yee K. Wong. PS-NeRF: Neural Inverse Rendering for Multi-view Photometric Stereo.

[24] **[ECCV 2022]** Xiaoming Li, **Chaofeng Chen**, Xianhui Lin, Wangmeng Zuo, Lei Zhang. From Face to Natural Image: Learning Real Degradation for Blind Image Super-Resolution.

[25] **[MM 2022; Oral]** Liang Liao, Kangmin Xu, Haoning Wu, **Chaofeng Chen**, Wenxiu Sun, Qiong Yan, Weisi Lin. Exploring the Effectiveness of Video Perceptual Representation in Blind Video Quality Assessment.

[26] **[CVPR 2021] Chaofeng Chen**, Xiaoming Li, Lingbo Yang, Xianhui Lin, Lei Zhang, Kwan-Yee K. Wong. Progressive Semantic-Aware Style Transformation for Blind Face Restoration.

[27] **[ICCV 2021]** Guanying Chen, **Chaofeng Chen**, Shi Guo, Zhetong Liang, Kwan-Yee K. Wong, Lei Zhang. HDR Video Reconstruction: A Coarse-to-fine Network and A Real-world Benchmark Dataset.

[28] **[ECCV 2020]** Xiaoming Li, **Chaofeng Chen**, Shangchen Zhou, Xianhui Lin, Wangmeng Zuo, Lei Zhang. Blind Face Restoration via Deep Multi-scale Component Dictionaries.

[29] **[ACCV 2018] Chaofeng Chen**, Wei Liu, Xiao Tan, Kwan-Yee K. Wong. Semi-Supervised Learning for Face Sketch Synthesis in the Wild.

[30] **[WACV 2018] Chaofeng Chen**[*], Xiao Tan[*], Kwan-Yee K. Wong. Face Sketch Synthesis with Style Transfer using Pyramid Column Feature.

[31] **[AAAI 2018; Oral]** Wei Liu, **Chaofeng Chen**, Kwan-Yee K. Wong. Char-Net: A Character-Aware Neural Network for Distorted Scene Text Recognition.

Selected Journal Papers

[32] **[TIP 2024; ESI Highly Cited Paper] Chaofeng Chen**, Jiadi Mo, Jingwen Hou, Haoning Wu, Liang Liao, Wenxiu Sun, Qiong Yan, Weisi Lin. TOPIQ: A Top-down Approach from Semantics to Distortions for Image Quality Assessment.

[33] **[CVIU 2023] Chaofeng Chen**, Wei Liu, Xiao Tan, Kwan-Yee K. Wong. Semi-supervised Cycle-GAN for face photo-sketch translation in the wild.

[34] **[TPAMI 2023]** Haoning Wu, **Chaofeng Chen**, Liang Liao, Jingwen Hou, Wenxiu Sun, Qiong Yan, Jinwei Gu, Weisi Lin. Neighbourhood Representative Sampling for Efficient End-to-end Video Quality Assessment.

[35] **[TCSVT 2023]** Haoning Wu, **Chaofeng Chen**, Liang Liao, Jingwen Hou, Wenxiu Sun, Qiong Yan, Weisi Lin. DisCoVQA: Temporal Distortion-Content Transformers for Video Quality Assessment.

[36] **[TIP 2023]** Wenqi Yang, Zhenfang Chen, **Chaofeng Chen**, Guanying Chen, Kwan-Yee K. Wong. Deep Face Video Inpainting via UV Mapping.

[37] **[TIP 2021] Chaofeng Chen**, Dihong Gong, Hao Wang, Zhifeng Li, Kwan-Yee K. Wong. Learning Spatial Attention for Face Super-Resolution.

Preprints

[38] **[arXiv preprint 2025]** Mengtian Li, Yunshu Bai, Yimin Chu, Yijun Shen, Zhongmei Li, Weifeng Ge, Zhifeng Xie, **Chaofeng Chen**[#]. GaussianMorphing: Mesh-Guided 3D Gaussians for Semantic-Aware Object Morphing.

[39] **[arXiv preprint 2025]** Hongyu Li, **Chaofeng Chen**, Xiaoming Li, Guangming Lu. 2D Gaussian Splatting with Semantic Alignment for Image Inpainting.

ACADEMIC ACTIVITIES

Area Chair: ICLR 2027

Senior Program Committee Member: AAAI 2027

Forums:

- PRCV 2026: Invited forum talk on quality assessment and reward modeling for generative vision (Aug. 2026).
- CCIG 2026: Forum Chair, Low-Quality Visual Processing and Quality Assessment (May 2026).

Committee Member:

- Member, Multimedia Technical Committee, China Society of Image and Graphics (CSIG).
- Executive Member, Multimedia Technical Committee, China Computer Federation (CCF).

Conference Reviewer: CVPR, ICCV, ECCV, SIGGRAPH Asia, ACM MM, ICLR, NeurIPS, ICML, AAAI.

Journal Reviewer: TPAMI, IJCV, TIP, CVIU, TIFS, TNNLS, Neurocomputing, TMM, TCSVT.

TEACHING

WHU | Spring 2026	Introduction to Machine Learning	Lecturer
SYSU | Spring 2025	AI236 Principles of Artificial Intelligence; AI238 Experiment of Artificial Intelligence	Lecturer

HONORS AND AWARDS

World's Top 2% Scientists, Stanford University	2025, 2026
ICML Silver Reviewer	2026
Hong Kong PhD Fellowship	2015 - 2018
National Scholarship	2013 - 2014